Ten Layers of the Annular-Norm and Complement-Mass Frontier

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Abstract

RH-302–RH-311 audit the direct annular reopening route and its interaction with the typed counterloop route. The slope-four annular tail is already closed, so the direct obstruction is exactly the moving head. Any positive annular theorem forces fixed-order head moments and must respect a sharp mass-limited rate barrier. The barrier is optimal for the coefficient envelope. On shrinking annuli the minimal-clock tail has an explicit threshold, and Hardy embedding gives the optimal square-root conversion loss. At the endpoint the actual mismatch belongs to H^2 but not H^∞; endpoint H^2 convergence remains possible only at a slow logarithmic rate. The typed alternative encounters a first alias exactly at the Gaussian localization horizon and requires a joint boundary-layer trace law. Neither route is activated. Both five-entry ledgers remain four of five, the cross-branch glue bit is false, and Gates A–E remain open.

1 Direct annular route

RH-302 proves that for every fixed $1.4 < \rho < \rho_*$,

$$\|g_\sigma - g_\sigma^{<m_\sigma}\|_{H^\infty(\rho)} + \|g_\sigma - g_\sigma^{<m_\sigma}\|_{H^2(\rho)} \longrightarrow 0,$$

where $m_\sigma = \lceil 4\log(1/\sigma)\rceil$. Thus the full norm converges if and only if the moving polynomial head does. RH-303 shows that such an aggregate theorem would bypass root pairing but would still force every fixed noisy-head moment to follow the counterloop, using the archived fixed-order total-trace and shell limits.

RH-304 and RH-305 quantify the mass cost of the exact odd anchors. Relative matching at the minimal bridge clock requires the mass exponent

$$0.9468615163684616\ldots,$$

and every actual modulus complement satisfies, on a strict annulus,

$$\|g_\sigma\|_{H^\infty/H^2(\rho)} \geq C_\rho \frac{M_\sigma^{-\kappa(\rho)}}{\log(e+M_\sigma)} \geq C'_\rho \frac{\sigma^{\kappa(\rho)}}{\log(e+1/\sigma)}, \qquad \kappa(\rho) = \frac{\log(1/(q_*\rho))}{\log(q_*/q)}.$$

At $\rho = 1.41$, $\kappa = 0.0350457053\ldots$. RH-306 proves that the power and logarithmic factor are sharp for the coefficient-envelope information class, while explicitly declining a spectral power-sum realization.

2 Shrinking gap and endpoint

RH-307 studies $\rho_\sigma = Re^{\eta_\sigma}$ at the minimal clock. If $\eta_\sigma = c\log L_\sigma/L_\sigma$, the mass-and-cap tail threshold is

$$c_* = \log(10/7) = 0.3566749439\ldots.$$

RH-308 then replaces the Cauchy conversion loss $\Theta(\eta^{-1})$ by the optimal Hardy loss $\Theta(\eta^{-1/2})$.

At $\rho_* = 1.426787483864073\ldots$, RH-309 proves the exact membership split

$$g_\sigma \in H^2(\rho_*), \qquad g_\sigma \notin H^\infty(\rho_*).$$

Endpoint H^2 convergence would close the full $R = 1.4$ coefficient budget with constant $4.9921110686\ldots$, but it must obey

$$\|g_\sigma\|_{H^2(\rho_*)} \geq C\{\log(e + M_\sigma)\}^{-1/2} \geq C'\{\log(e + 1/\sigma)\}^{-1/2}.$$

This tends to zero and therefore leaves endpoint convergence open.

3 Typed first-alias route

RH-310 ties the counterloop period to the endpoint-resolution rank. The first alias order and the old localization horizon have the same slope $1/\log\lambda$. Its cycle clearance is $\Theta(\sigma)$, and the alias and parity weighted exponents coincide at $0.4634069445\ldots$. Minimal-clock convergence would require

$$c_{\sigma,2k} - c_{2k} = (2k-2)\beta_k^{2k} + 2\beta^{2k} + o(kR^{-2k}).$$

No archived theorem supplies this joint moving-order boundary-layer law.

4 Typed frontier

branch	head	bridge	tail	target	boundary	score
noisy modulus spectrum	1	0	1	1	1	4
graded monodromy counterloop	1	1	0	1	1	4

The weighted cross-branch glue remains false and the complete count is zero. No coordinatewise union is permitted because the two heads remain distinct.

Proposition 4.1 (two legitimate reopening inputs)**.** *The direct branch reopens if the actual mismatch converges in endpoint $H^2(\rho_*)$, or in an interior annular norm at a rate compatible with the mass barrier. The typed branch reopens only with a parity-renormalized, alias-inclusive joint boundary-layer trace estimate satisfying the exact first-alias matching law. More high-order tail estimates, pre-alias fixed moments, or separate absolute parity/shell majorants are not reopening inputs.*

Proof. RH-309 plus the Hardy coefficient inequality turns endpoint H^2 convergence directly into the full weighted complement bridge. RH-302 and RH-300 do the same for any fixed interior annulus. Otherwise RH-288 requires the two typed errors on a common clock, and RH-310 identifies their first unavoidable moving-order obligation. □

Remark 4.2. No Gate-A determinant identification has been completed. Gates B–E remain false/open. This batch constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace formula or zeta-divisor equality, and does not imply RH.